Fiscal Year:	FY 2015	**Task Last Updated:**	FY 04/27/2016
PI Name:	Baulch, Janet E. Ph.D.		
Project Title:	Space Radiation Effects on Genomic and Epigenetic Stability: Influence of Radiation Quality and Genetic Predisposition		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jbaulch@uci.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	949.824.7396
Organization Name:	University of California, Irvine		
PI Address 1:	Department of Radiation Oncology		
PI Address 2:	Medical Sciences I, B149		
PI Web Page:			
City:	Irvine	**State:**	CA
Zip Code:	92697	**Congressional District:**	45
Comments:	NOTE: PI previously at University of Maryland; moved to UCA Irvine in summer 2013 (Ed., 11/11/13)		
Project Type:	Ground	**Solicitation / Funding Source:**	2010 Space Radiobiology NNJ10ZSA001N
Start Date:	07/30/2013	**End Date:**	07/29/2016
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
Contact Email:	lisa.c.simonsen@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Extended to 7/29/2016 per NSSC information (Ed., 7/14/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX13AK70G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	[Editor's note 11/11/2013. Note this is a continuation of "Space Radiation Effects on Genomic and Epigenetic Stability: Influence of Radiation Quality and Genetic Predisposition" (grant NNX11AC55G); see that project for previous reports.] Astronauts traveling in space are exposed to a variety of radiations. It is known that exposure to ionizing radiation can lead to an increased risk for developing cancer. Since there is currently no way to protect astronauts, it is important that the carcinogenic risk of exposures to these radiations is understood. The delayed, deleterious effects of radiation exposure are grouped under the terms radiation-induced genomic and epigenomic instability. These mechanisms are considered to be early steps in the carcinogenic process. Epigenetic mechanisms are thought to play a significant role in carcinogenesis by regulating gene expression and genome stability. These epigenetic mechanisms include DNA methylation, miRNA mediated gene silencing, and histone modification. There are also genetic factors in mouse and		

	humans that determine the relative sensitivity of their cells to radiation-induced effects and ultimately their cancer risk. This project will develop assays that will provide functional biomarkers for exposure that are also indicators of carcinogenic risk. We will accomplish these objectives by evaluating mice with differing sensitivities to radiation for epigenomic instability as measured by changes in miRNA and DNA methylation profiles following exposure to iron, silicon, or oxygen ion radiations. The effect of age on the epigenetic radiation induced response will be evaluated using mice of relatively younger and older ages.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The goal of this grant is to evaluate the effect of space radiation exposure on epigenetic profiles with a focus on the role for genetic factors, age, and radiation linear energy transfer (LET). Some of the research and technology questions that this proposal addresses include the following: 1) the probabilities for increased carcinogenic risk following relatively low dose radiation exposure as a function of NASA's operational parameters (radiation quality, age at exposure, genetic factors, tissue specific response); 2) the development of quantitative, functional biomarkers for carcinogenic risk; 3) a better understanding of the molecular mechanisms involved in cancer initiation and progression. Understanding the cancer risk associated with exposure to either terrestrial or space radiation is important for human health. Epigenetic abnormalities are thought to play a key role in the earliest steps of carcinogenic initiation. Epigenetic changes are mitotically and meiotically heritable, stable alteration in gene expression that include DNA methylation, histone modifications, and RNA associated gene silencing. This project tests the hypothesis that radiation exposure can affect epigenetic profiles and increase carcinogenic risk. We further hypothesize that genetic factors and age at the time of exposure impact the epigenetic radiation response. Specific Aim 1 will evaluate radiation-induced alterations in miR profiles and gene expression. For each tissue, dose, and time point, genome-wide miR profiles will be analyzed. In parallel, global mRNA profiles will be analyzed to provide information regarding gene expression. Together these arrays will allow us to identify up-regulated miRNA that correlate with down-regulated target mRNA, or vis versa. Mining these data will allow us to identify affected pathways that are relevant to cancer and the radiation response. Specific aim 2 will evaluate the effect of irradiation on the induction of aberrant DNA methylation. Alterations in DNA methylation have emerged as one of the most consistent molecular markers in multiple cancers and for carcinogenic initiation and progression. We will measure genome wide methylation by evaluating repeat element methylation. These repeat elements make up a significant portion of the mammalian genome and aberrant repeat element methylation has been linked to genomic instability, altered gene expression, and cancer. We will also evaluate promoter methylation for genes associated with increased carcinogenic risk and the radiation response. Using these three strategies to evaluate tissue specific DNA methylation in irradiated mice we will be able to survey genome wide changes that have been linked to carcinogenic processes. As a whole, this project uses a systems biology approach that will provide information regarding tissue-specific effects of radiation LET, genetic factors, age, and post-irradiation time on miRNA-mediated gene regulation as it applies to the radiation response and radiation-induced carcinogenesis. This benefits not only our astronauts, but those of us on Earth, by increasing our understanding of the molecular mechanisms of cancer and the radiation response.
Task Progress:	[Ed. note 4/2016: Report submission delayed; covers period July 2014 through July 2015.] In collaboration with the laboratories of Dr. Munjal Acharya and Dr. Charles Limoli this project has also expanded to evaluate levels of DNA methylation in the brain of irradiated mice. DNA methylation has been classically defined as 5-methylcytosine (5mC) at a CpG dinucleotide sequence. Recently additional cytosine methylation modifications have been identified; particularly 5hmC has been implicated in CNS-specific epigenetic gene regulation. Both 5mC and 5hmC play key roles in gene expression that can affect changes in neuronal signaling, altered synaptic plasticity, cognition and neurodegeneration. Preliminary data demonstrate that global 5mC and 5hmC levels are increased as early as 24 hours after exposure to 20 cGy of Si particles and persist for as long as a month post exposure. The same significant increases are also observed following exposure to low LET x-rays. We have also performed 5mC and 5hmC DNA methylation arrays that demonstrate highly significant changes in the levels of both types of cytosine modifications in the hippocampus of the irradiated mouse. Among the hits obtained from these arrays are genes involved in synaptic plasticity and neuronal signaling. Ongoing RT-PCR experiments will evaluate the functionality of these changes in DNA methylation. While the 5mC modifications are effected by DNA methyltransferases, 5mC is converted to 5hmC by the TET family of enzymes. We have now evaluated TET1 protein levels in the brain of mice irradiated using 20 cGy of 28Si particles. These significant increases in TET1 suggest that changes in DNA methylation in the brain of the irradiated mouse are part of an active, enzymatically driven change, rather than an indirect, oxidative stress mediated response to irradiation. Ongoing experiments will further elucidate this important CNS radiation response. During this final year, my laboratory will complete DNA methylation data for the C57Bl/6J and C3H/HeN mice exposed to 56Fe and 28Si ions for comparison of dose, radiation LET, time post exposure, and genetic variability on DNA methylation levels. Additional collaborative work with Drs. Acharya and Limoli will complete the evaluation of DNA methylation in the brain of the irradiated mice in an effort to link DNA methylation, DNA methylating enzyme levels/activity, and radiation induced cognitive deficits. Some of the above-described studies have been presented at the 2015 NASA Space Radiation Investigators' Workshop in Galveston (Posters; Baulch, Acharya), the Heavy Ions in Therapy and Space Satellite meeting in Osaka, Japan (Oral Presentation; Baulch), or the International Congress for Radiation Research in Kyoto, Japan (Poster; Acharya). Additional work will be presented at the 2015 Radiation Research Society meeting in Westin, FL in September, 2015 (Poster; Acharya) and the 2015 Environmental Mutagenesis and Genomics Society meeting in New Orleans, LA in September, 2015 (Poster; Baulch).
Bibliography Type:	Description: (Last Updated: 05/01/2017)

Articles in Peer-reviewed Journals	Lasio G, Guerrero M, Goetz W, Lima F, Baulch JE. "Effect of varying dose-per-pulse and average dose rate in X-ray beam irradiation on cultured cell survival." Radiat Environ Biophys. 2014 Nov;53(4):671-6. http://dx.doi.org/10.1007/s00411-014-0565-2 ; PubMed PMID: 25169705 , Nov-2014
Articles in Peer-reviewed Journals	Baulch JE, Aypar U, Waters KM, Yang AJ, Morgan WF. "Genetic and epigenetic changes in chromosomally stable and unstable progeny of irradiated cells." PLoS One. 2014 Sep 24;9(9):e107722. eCollection 2014. http://dx.doi.org/10.1371/journal.pone.0107722 ; PubMed PMID: 25251398; PubMed Central PMCID: PMC4175465 , Sep-2014
Articles in Peer-reviewed Journals	Keszenman DJ, Kolodiuk L, Baulch JE. "DNA damage in cells exhibiting radiation-induced genomic instability." Mutagenesis. 2015 May;30(3):451-8. Epub 2015 Feb 22. http://dx.doi.org/10.1093/mutage/gev006 ; PubMed PMID: 25711497 , May-2015
Articles in Peer-reviewed Journals	Baulch JE, Craver BM, Tran KK, Yu L, Chmielewski N, Allen BD, Limoli CL. "Persistent oxidative stress in human neural stem cells exposed to low fluences of charged particles." Redox Biol. 2015 Aug;5:24-32. Epub 2015 Mar 11. http://dx.doi.org/10.1016/j.redox.2015.03.001 ; PubMed PMID: 25800120; PubMed Central PMCID: PMC4371546 , Aug-2015
Articles in Peer-reviewed Journals	Parihar VK, Allen B, Tran KK, Macaraeg TG, Chu EM, Kwok SF, Chmielewski NN, Craver BM, Baulch JE, Acharya MM, Cucinotta FA, Limoli CL. "What happens to your brain on the way to Mars." Sci Adv. 2015 May 1;1(4): e1400256. http://dx.doi.org/10.1126/sciadv.1400256 , May-2015